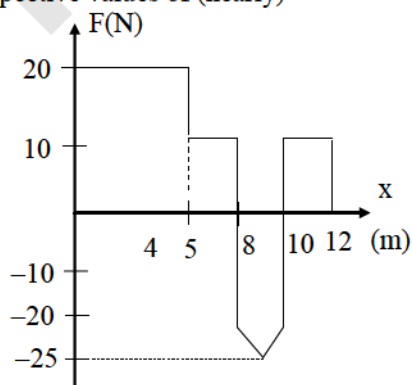


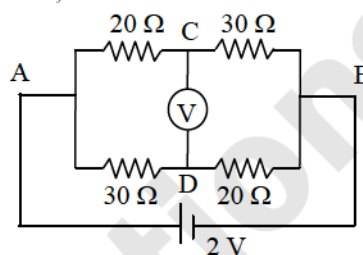
NEET (UG) - 2019 (Odisha) Question Paper

- A sphere encloses an electric dipole with charges $\pm 3 \times 10^{-6} \text{ C}$. What is the total electric flux across the sphere?
(A) $6 \times 10^{-6} \text{ Nm}^2/\text{C}$ (B) $-3 \times 10^{-6} \text{ Nm}^2/\text{C}$
(C) Zero (D) $3 \times 10^{-6} \text{ Nm}^2/\text{C}$
- The main scale of a vernier callipers has n divisions/cm. n divisions of the vernier scale coincide with $(n - 1)$ divisions of main scale. The least count of the vernier callipers is,
(A) $\frac{1}{n(n+1)} \text{ cm}$ (B) $\frac{1}{(n+1)(n-1)} \text{ cm}$
(C) $\frac{1}{n} \text{ cm}$ (D) $\frac{1}{n^2} \text{ cm}$
- An LED is constructed from a p-n junction diode using GaAsP. The energy gap is 1.9 eV. The wavelength of the light emitted will be equal to
(A) $654 \times 10^{-11} \text{ m}$ (B) $10.4 \times 10^{-26} \text{ m}$
(C) 654 nm (D) 654 Å
- $(20\hat{i} + 25\hat{j} - 12\hat{k})$ suddenly breaks into two pieces whose masses are in the ratio 1 : 5. The smaller mass flies off with a velocity $(100\hat{i} + 35\hat{j} + 8\hat{k})$. The velocity of the larger piece will be,
(A) $-20\hat{i} - 15\hat{j} - 80\hat{k}$ (B) $4\hat{i} + 23\hat{j} - 16\hat{k}$
(C) $-100\hat{i} - 35\hat{j} - 8\hat{k}$ (D) $20\hat{i} + 15\hat{j} - 80\hat{k}$
- Two toroid's 1 and 2 have total number of turns 200 and 100 respectively with average radii 40 cm and 20 cm respectively. If they carry same current I , the ratio of the magnetic fields along the two loops is,
(A) 1 : 2 (B) 1 : 1 (C) 4 : 1 (D) 2 : 1
- An object of mass 500 g, initially at rest, is acted upon by a variable force whose X-component varies with X in the manner shown. The velocities of the object at the points $X = 8 \text{ m}$ and $X = 12 \text{ m}$, would have the respective values of (nearly)



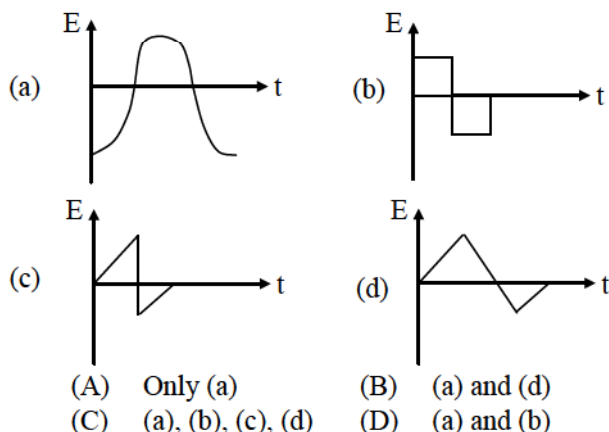
- (A) 18 m/s and 20.6 m/s
(B) 18 m/s and 24.4 m/s
(C) 23 m/s and 24.4 m/s
(D) 23 m/s and 20.6 m/s

- The reading of an ideal voltmeter in the circuit shown is,



- (A) 0.4 V (B) 0.6 V
(C) 0 V (D) 0.5 V

- A solid cylinder of mass 2 kg and radius 50 cm rolls up an inclined plane of angle of inclination 30° . The centre of mass of the cylinder has speed of 4 m/s. The distance travelled by the cylinder on the inclined surface will be, [take $g = 10 \text{ m/s}^2$]
(A) 2.4 m (B) 2.2 m
(C) 1.6 m (D) 1.2 m
- In a Young's double slit experiment, if there is no initial phase difference between the light from the two slits, a point on the screen corresponding to the fifth minimum has path difference
(A) $11 \frac{\lambda}{2}$ (B) $5 \frac{\lambda}{2}$
(C) $10 \frac{\lambda}{2}$ (D) $9 \frac{\lambda}{2}$
- The variation of e.m.f. with time for four types of generators are shown in the figures. Which amongst them can be called AC?



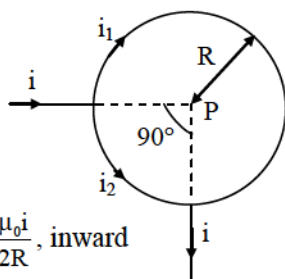
- (A) Only (a) (B) (a) and (d)
(C) (a), (b), (c), (d) (D) (a) and (b)



11. The time period of a geostationary satellite is 24 h, at a height $6R_E$ (R_E is radius of earth) from surface of earth. The time period of another satellite whose height is $2.5 R_E$ from surface will be,

(A) $\frac{12}{2.5} h$ (B) $6\sqrt{2} h$
(C) $12\sqrt{2} h$ (D) $\frac{24}{2.5} h$

12. A straight conductor carrying current i splits into two parts as shown in the figure. The radius of the circular loop is R . The total magnetic field at the centre P of the loop is,

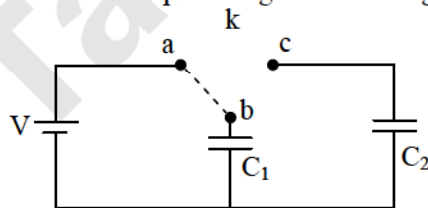


(A) $\frac{\mu_0 i}{2R}$, inward
(B) Zero
(C) $3\mu_0 i/32R$, outward
(D) $3\mu_0 i/32R$, inward

13. Assuming that the gravitational potential energy of an object at infinity is zero, the change in potential energy (final – initial) of an object of mass m , when taken to a height h from the surface of earth (of radius R), is given by,

(A) $\frac{GMm}{R+h}$ (B) $-\frac{GMm}{R+h}$
(C) $\frac{GMmh}{R(R+h)}$ (D) mgh

14. Two identical capacitors C_1 and C_2 of equal capacitance are connected as shown in the circuit. Terminals a and b of the key k are connected to charge capacitor C_1 using battery of e. m. f. V volt. Now disconnecting a and b the terminals b and c are connected. Due to this, what will be the percentage loss of energy?



(A) 25% (B) 75%
(C) 0% (D) 50%

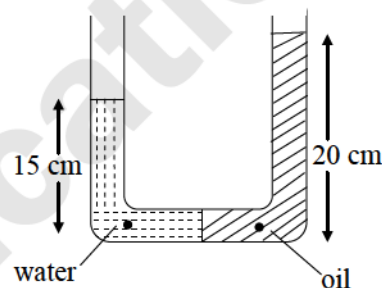
15. The stress-strain curves are drawn for two different materials X and Y. It is observed that the ultimate strength point and the fracture point are close to each other for material X but are far apart for material Y. We can say that materials X and Y are likely to be (respectively),

(A) Plastic and ductile
(B) Ductile and brittle
(C) Brittle and ductile
(D) Brittle and plastic

16. A cycle wheel of radius 0.5 m is rotated with constant angular velocity of 10 rad/s in a region of magnetic field of 0.1 T which is perpendicular to the plane of the wheel. The e.m.f. generated between its centre and the rim is,

(A) Zero (B) 0.25 V
(C) 0.125 V (D) 0.5 V

17. In a u-tube as shown in the fig. water and oil are in the left side and right side of the tube respectively. The heights from the bottom for water and oil columns are 15 cm and 20 cm respectively. The density of the oil is [take $\rho_{\text{water}} = 1000 \text{ kg/m}^3$]



(A) 1333 kg/m^3 (B) 1200 kg/m^3
(C) 750 kg/m^3 (D) 1000 kg/m^3

18. A particle of mass $5m$ at rest suddenly breaks on its own into three fragments. Two fragments of mass m each move along mutually perpendicular direction with speed v each. The energy released during the process is,

(A) $\frac{4}{3}mv^2$ (B) $\frac{3}{5}mv^2$
(C) $\frac{5}{3}mv^2$ (D) $\frac{3}{2}mv^2$

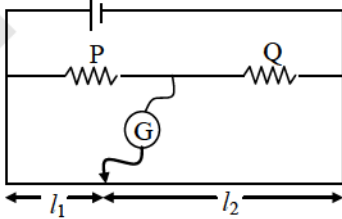
19. An object kept in a large room having air temperature of 25°C takes 12 minutes to cool from 80°C to 70°C . The time taken to cool for the same object from 70°C to 60°C would be nearly,

(A) 15 min (B) 10 min
(C) 12 min (D) 20 min

20. The value of $\gamma \left(= \frac{C_p}{C_v} \right)$ for hydrogen, helium and another ideal diatomic gas X (whose molecules are not rigid but have an additional vibrational mode), are respectively equal to,

(A) $\frac{7}{5}, \frac{5}{3}, \frac{7}{4}$ (B) $\frac{7}{5}, \frac{5}{3}, \frac{9}{7}$
(C) $\frac{5}{3}, \frac{7}{5}, \frac{9}{7}$ (D) $\frac{5}{3}, \frac{7}{5}, \frac{7}{5}$



21. Two bullets are fired horizontally and simultaneously towards each other from roof tops of two buildings 100 m apart and of same height of 200 m, with the same velocity of 25 m/s. When and where will the two bullets collide? ($g = 10 \text{ m/s}^2$)
 (A) They will not collide
 (B) After 2 s at a height of 180 m
 (C) After 2 s at a height of 20 m²
 (D) After 4 s at a height of 120 m
22. A tuning fork with frequency 800 Hz produces resonance in a resonance column tube with upper end open and lower end closed by water surface. Successive resonance are observed at lengths 9.75 cm, 31.25 cm and 52.75 cm. The speed of sound in air is,
 (A) 172 m/s (B) 500 m/s
 (C) 156 m/s (D) 344 m/s
23. Two metal spheres, one of radius R and the other of radius $2R$ respectively have the same surface charge density σ . They are brought in contact and separated. What will be the new surface charge densities on them?
 (A) $\sigma_1 = \frac{5}{3}\sigma$, $\sigma_2 = \frac{5}{6}\sigma$ (B) $\sigma_1 = \frac{5}{6}\sigma$, $\sigma_2 = \frac{5}{2}\sigma$
 (C) $\sigma_1 = \frac{5}{2}\sigma$, $\sigma_2 = \frac{5}{6}\sigma$ (D) $\sigma_1 = \frac{5}{2}\sigma$, $\sigma_2 = \frac{5}{3}\sigma$
24. A body of mass m is kept on a rough horizontal surface (coefficient of friction = μ). A horizontal force is applied on the body, but it does not move. The resultant of normal reaction and the frictional force acting on the object is given by F , where F is,
 (A) $|\vec{F}| = mg$ (B) $|\vec{F}| = mg + \mu mg$
 (C) $|\vec{F}| = \mu mg$ (D) $|\vec{F}| \leq mg \sqrt{1 + \mu^2}$
25. The metre bridge shown is in balance position with $\frac{P}{Q} = \frac{l_1}{l_2}$. If we now interchange the positions of galvanometer and cell, will the bridge work? If yes, what will be balance condition?
- 
- (A) yes, $\frac{P}{Q} = \frac{l_1}{l_2}$ (B) yes, $\frac{P}{Q} = \frac{l_2 - l_1}{l_2 + l_1}$
 (C) no, no null point (D) yes, $\frac{P}{Q} = \frac{l_2}{l_1}$
26. The relations amongst the three elements of earth's magnetic field, namely horizontal component H , vertical component V and dip δ are, (B_E = total magnetic field)
 (A) $V = B_E$, $H = B_E \tan \delta$
 (B) $V = B_E \tan \delta$, $H = B_E$
 (C) $V = B_E \sin \delta$, $H = B_E \cos \delta$
 (D) $V = B_E \cos \delta$, $H = B_E \sin \delta$
27. A deep rectangular pond of surface area A , containing water (density = ρ , specific heat capacity = s), is located in a region where the outside air temperature is at a steady value of -26°C . The thickness of the frozen ice layer in this pond, at a certain instant is x . Taking the thermal conductivity of ice as K , and its specific latent heat of fusion as L , the rate of increase of the thickness of ice layer, at this instant, would be given by
 (A) $26K/\rho x(L + 4s)$ (B) $26K/\rho x(L - 4s)$
 (C) $26K/(\rho x^2 L)$ (D) $26K/(\rho x L)$
28. A circuit when connected to an AC source of 12 V gives a current of 0.2 A. The same circuit when connected to a DC source of 12 V, gives a current of 0.4 A. The circuit is
 (A) Series LCR (B) Series LR
 (C) Series RC (D) Series LC
29. For a transparent medium, relative permeability and permittivity, μ_r and ϵ_r are 1.0 and 1.44 respectively. The velocity of light in this medium would be,
 (A) $4.32 \times 10^8 \text{ m/s}$ (B) $2.5 \times 10^8 \text{ m/s}$
 (C) $3 \times 10^8 \text{ m/s}$ (D) $2.08 \times 10^8 \text{ m/s}$
30. A truck is stationary and has a bob suspended by a light string, in a frame attached to the truck. The truck suddenly moves to the right with an acceleration of a . The pendulum will tilt
 (A) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{g}{a}\right)$
 (B) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{g}{a}\right)$
 (C) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}\left(\frac{a}{g}\right)$
 (D) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}\left(\frac{a}{g}\right)$
31. A double convex lens has focal length 25 cm. The radius of curvature of one of the surfaces is double of the other. Find the radii if the refractive index of the material of the lens is 1.5.
 (A) 50 cm, 100 cm
 (B) 100 cm, 50 cm
 (C) 25 cm, 50 cm
 (D) 18.75 cm, 37.5 cm



32. A particle starting from rest, moves in a circle of radius 'r'. It attains a velocity of V_0 m/s in the n^{th} round. Its angular acceleration will be,
 (A) $\frac{V_0^2}{4\pi nr}$ rad/s² (B) $\frac{V_0}{n}$ rad/s²
 (C) $\frac{V_0^2}{2\pi nr^2}$ rad/s² (D) $\frac{V_0^2}{4\pi nr^2}$ rad/s²
33. Angular width of the central maxima in the Fraunhofer diffraction for $\lambda = 6000 \text{ \AA}$ is θ_0 . When the same slit is illuminated by another monochromatic light, the angular width decreases by 30%. The wavelength of this light is
 (A) 420 $\text{\AA}$ (B) 1800 $\text{\AA}$
 (C) 4200 $\text{\AA}$ (D) 6000 $\text{\AA}$
34. 1 g of water, of volume 1 cm³ at 100°C, is converted into steam at same temperature under normal atmospheric pressure ($\approx 1 \times 10^5 \text{ Pa}$). The volume of steam formed equals 1671 cm³. If the specific latent heat of vaporisation of water is 2256 J/g, the change in internal energy is,
 (A) 2256 J (B) 2423 J
 (C) 2089 J (D) 167 J
35. The work function of a photosensitive material is 4.0 eV. The longest wavelength of light that can cause photon emission from the substance is (approximately)
 (A) 310 nm (B) 3100 nm
 (C) 966 nm (D) 31 nm
36. A person standing on the floor of an elevator drops a coin. The coin reaches the floor in time t_1 if the elevator is at rest and in time t_2 if the elevator is moving uniformly. Then
 (A) $t_1 = t_2$
 (B) $t_1 < t_2$ or $t_1 > t_2$ depending upon whether the lift is going up or down
 (C) $t_1 < t_2$
 (D) $t_1 > t_2$
37. A proton and an α -particle are accelerated from rest to the same energy. The de Broglie wavelengths λ_p and λ_α are in the ratio,
 (A) 4 : 1 (B) 2 : 1
 (C) 1 : 1 (D) $\sqrt{2} : 1$
38. A mass falls from a height 'h' and its time of fall 't' is recorded in terms of time period T of a simple pendulum. On the surface of earth it is found that $t = 2T$. The entire set up is taken on the surface of another planet whose mass is half of that of earth and radius the same. Same experiment is repeated and corresponding times noted as t' and T' . Then we can say
 (A) $t' = 2T'$ (B) $t' = \sqrt{2} T'$
 (C) $t' > 2T'$ (D) $t' < 2T'$
39. The radius of the first permitted Bohr orbit, for the electron, in a hydrogen atom equals 0.51 $\text{\AA}$ and its ground state energy equals -13.6 eV. If the electron in the hydrogen atom is replaced by muon (μ^-) [charge same as electron and mass 207 m_e], the first Bohr radius and ground state energy will be,
 (A) $2.56 \times 10^{-13} \text{ m}$, -13.6 eV
 (B) $0.53 \times 10^{-13} \text{ m}$, -3.6 eV
 (C) $25.6 \times 10^{-13} \text{ m}$, -2.8 eV
 (D) $2.56 \times 10^{-13} \text{ m}$, -2.8 keV
40. Two small spherical metal balls, having equal masses, are made from materials of densities ρ_1 and ρ_2 ($\rho_1 = 8\rho_2$) and have radii of 1 mm and 2 mm, respectively, they are made to fall vertically (from rest) in a viscous medium whose coefficient of viscosity equals η and whose density is $0.1\rho_2$. The ratio of their terminal velocities would be,
 (A) $\frac{79}{36}$ (B) $\frac{79}{72}$ (C) $\frac{19}{36}$ (D) $\frac{39}{72}$
41. The rate of radioactive disintegration at an instant for a radioactive sample of half-life $2.2 \times 10^9 \text{ s}$ is 10^{10} s^{-1} . The number of radioactive atoms in that sample at that instant is,
 (A) 3.17×10^{19} (B) 3.17×10^{20}
 (C) 3.17×10^{17} (D) 3.17×10^{18}
42. The distance covered by a particle undergoing SHM in one time period is (amplitude = A),
 (A) 4A (B) Zero (C) A (D) 2A
43. The circuit diagram shown here corresponds to the logic gate,

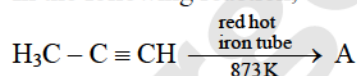
 (A) NAND (B) NOR
 (C) AND (D) OR
44. A person travelling in a straight line moves with a constant velocity v_1 for certain distance 'x' and with a constant velocity v_2 for next equal distance. The average velocity v is given by the relation
 (A) $v = \sqrt{v_1 v_2}$ (B) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2}$
 (C) $\frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$ (D) $\frac{v}{2} = \frac{v_1 + v_2}{2}$



45. An equiconvex lens has power P . It is cut into two symmetrical halves by a plane containing the principal axis. The power of one part will be,
(A) P (B) 0 (C) $\frac{P}{2}$ (D) $\frac{P}{4}$
46. In water saturated air, the mole fraction of water vapour is 0.02 . If the total pressure of the saturated air is 1.2 atm, the partial pressure of dry air is _____.
(A) 0.98 atm (B) 1.18 atm
(C) 1.76 atm (D) 1.176 atm
47. The amine that reacts with Hinsberg reagent to give an alkali insoluble product is _____.
(A) $\text{CH}_3 - \underset{\text{H}_3\text{C}}{\overset{\text{CH}_3}{\text{C}}} - \text{CH} - \text{NH}_2$
(B) $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{NH} - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_3$
(C) $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_2\text{CH}_3}{\text{N}} - \text{CH}_2\text{CH}_3$
(D) $\text{CH}_3 - \underset{\text{CH}_3}{\overset{\text{NH}_2}{\text{C}}} - \text{CH}_2\text{CH}_2\text{CH}_3$

48. The volume occupied by 1.8 g of water vapour at 374°C and 1 bar pressure will be _____.
[Use $R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$]
(A) 5.37 L (B) 96.66 L
(C) 55.87 L (D) 3.10 L

49. In the following reaction,



The number of sigma (σ) bonds present in the product A, is _____.

- (A) 18 (B) 21 (C) 9 (D) 24
50. Which of the following compounds is used in cosmetic surgery?
(A) Zeolites (B) Silica
(C) Silicates (D) Silicones
51. The major products C and D formed in the following reaction respectively are _____.
 $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{C}(\text{CH}_3)_3 \xrightarrow[\Delta]{\text{excess HI}} \text{C} + \text{D}$
(A) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{OH}$ and $\text{OH} - \text{C}(\text{CH}_3)_3$
(B) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{I}$ and $\text{I} - \text{C}(\text{CH}_3)_3$
(C) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{OH}$ and $\text{I} - \text{C}(\text{CH}_3)_3$
(D) $\text{H}_3\text{C} - \text{CH}_2 - \text{CH}_2 - \text{I}$ and $\text{HO} - \text{C}(\text{CH}_3)_3$

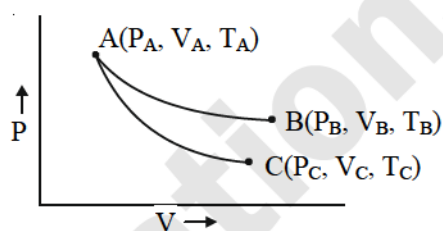
52. The artificial sweetener stable at cooking temperature and does NOT provide calories is _____.

(A) alitame (B) saccharin
(C) aspartame (D) sucralose

53. Orbital having 3 angular nodes and 3 total nodes is _____.

(A) $6d$ (B) $5p$
(C) $3d$ (D) $4f$

54. Reversible expansion of an ideal gas under isothermal and adiabatic conditions are as shown in the figure.



$AB \rightarrow$ Isothermal expansion

$AC \rightarrow$ Adiabatic expansion

Which of the following options is INCORRECT?

- (A) $T_C > T_A$
(B) $\Delta S_{\text{isothermal}} > \Delta S_{\text{adiabatic}}$
(C) $T_A = T_B$
(D) $W_{\text{isothermal}} > W_{\text{adiabatic}}$

55. When neutral or faintly alkaline KMnO_4 is treated with potassium iodide, iodide ion is converted into 'X'. 'X' is _____.
(A) IO^- (B) I_2 (C) IO_4^- (D) IO_3^-

56. A first order reaction has a rate constant of $2.303 \times 10^{-3} \text{ s}^{-1}$. The time required for 40 g of this reactant to reduce to 10 g will be _____.
[Given that $\log_{10} 2 = 0.3010$]
(A) 602 s (B) 230.3 s
(C) 301 s (D) 2000 s

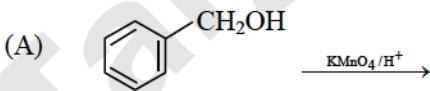
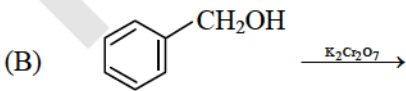
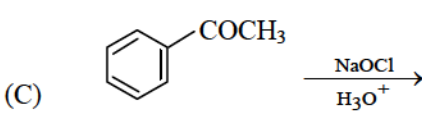
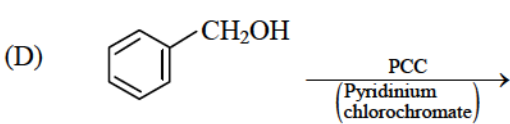
57. Which of the following CANNOT act both as Bronsted acid and as Bronsted base?

(A) HSO_4^- (B) HCO_3^-
(C) NH_3 (D) HCl

58. The standard electrode potential (E°) values of Al^{3+}/Al , Ag^+/Ag , K^+/K and Cr^{3+}/Cr are -1.66 V, 0.80 V, -2.93 V and -0.74 V, respectively. The CORRECT decreasing order of reducing power of the metal is _____.
(A) $\text{Al} > \text{K} > \text{Ag} > \text{Cr}$ (B) $\text{Ag} > \text{Cr} > \text{Al} > \text{K}$
(C) $\text{K} > \text{Al} > \text{Cr} > \text{Ag}$ (D) $\text{K} > \text{Al} > \text{Ag} > \text{Cr}$

59. The oxidation state of Cr in CrO_6 is _____.
(A) $+4$ (B) -6
(C) $+12$ (D) $+6$



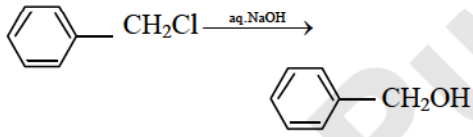
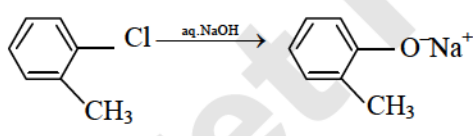
60. Which of the following statements is CORRECT regarding a solution of two components A and B exhibiting positive deviation from ideal behaviour?
- (A) Intermolecular attractive forces between A – A and B – B are equal to those between A – B.
 (B) Intermolecular attractive forces between A – A and B – B are stronger than those between A – B.
 (C) $\Delta_{\text{mix}}H = 0$ at constant T and P.
 (D) $\Delta_{\text{mix}}V = 0$ at constant T and P.
61. Identify the INCORRECT statement.
- (A) Gangue is an ore contaminated with undesired materials.
 (B) The scientific and technological process used for isolation of the metal from its ore is known as metallurgy.
 (C) Minerals are naturally occurring chemical substances in the earth's crust.
 (D) Ores are minerals that may contain a metal.
62. Which of the following oxoacids of phosphorus has the strongest reducing property?
- (A) H_3PO_4 (B) $\text{H}_4\text{P}_2\text{O}_7$
 (C) H_3PO_3 (D) H_3PO_2
63. Which of the following is the correct order of dipole moment?
- (A) $\text{H}_2\text{O} < \text{NF}_3 < \text{NH}_3 < \text{BF}_3$
 (B) $\text{NH}_3 < \text{BF}_3 < \text{NF}_3 < \text{H}_2\text{O}$
 (C) $\text{BF}_3 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
 (D) $\text{BF}_3 < \text{NH}_3 < \text{NF}_3 < \text{H}_2\text{O}$
64. In hydrogen atom, the de Broglie wavelength of an electron in the second Bohr orbit is _____.
 [Given that Bohr radius, $a_0 = 52.9 \text{ pm}$]
- (A) 105.8 pm (B) 211.6 pm
 (C) $211.6 \pi \text{ pm}$ (D) $52.9 \pi \text{ pm}$
65. The reaction that does NOT give benzoic acid as the major product is _____.
 (A)  $\xrightarrow{\text{KMnO}_4/\text{H}^+}$
 (B)  $\xrightarrow{\text{K}_2\text{Cr}_2\text{O}_7}$
 (C)  $\xrightarrow[\text{H}_3\text{O}^+]{\text{NaOCl}}$
 (D)  $\xrightarrow[\text{(Pyridinium chlorochromate)}]{\text{PCC}}$
66. The liquified gas that is used in dry cleaning along with a suitable detergent is _____.
 (A) CO_2 (B) Water gas
 (C) Petroleum gas (D) NO_2
67. The pH of 0.01 M $\text{NaOH}_{(\text{aq})}$ solution will be _____.
 (A) 9 (B) 7.01 (C) 2 (D) 12
68. The polymer that is used as a substitute for wool in making commercial fibres is _____.
 (A) buna-N (B) melamine
 (C) nylon-6,6 (D) polyacrylonitrile
69. Following limiting molar conductivities are given as,
 $\Lambda_m^\circ(\text{H}_2\text{SO}_4) = x \text{ S cm}^2 \text{ mol}^{-1}$
 $\Lambda_m^\circ(\text{K}_2\text{SO}_4) = y \text{ S cm}^2 \text{ mol}^{-1}$
 $\Lambda_m^\circ(\text{CH}_3\text{COOK}) = z \text{ S cm}^2 \text{ mol}^{-1}$
 Λ_m° (in $\text{S cm}^2 \text{ mol}^{-1}$) for CH_3COOH will be _____.
 (A) $\left(\frac{x-y}{2}\right) + z$ (B) $x - y + 2z$
 (C) $x + y + z$ (D) $x - y + z$
70. Aluminium chloride in acidified aqueous solution forms a complex 'A', in which hybridisation state of Al is 'B'. What are 'A' and 'B', respectively?
 (A) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, d^2sp^3
 (B) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$, sp^3d^2
 (C) $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$, sp^3
 (D) $[\text{Al}(\text{H}_2\text{O})_4]^{3+}$, dsp^2
71. The number of hydrogen bonded water molecule(s) associated with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is _____.
 (A) 5 (B) 3 (C) 1 (D) 2
72. Which of the alkali metal chloride (MCl) forms its dihydrate salt ($\text{MCl} \cdot 2\text{H}_2\text{O}$) easily?
 (A) KCl (B) LiCl
 (C) CsCl (D) RbCl
73. Match the catalyst with the process.

	Catalyst		Process
i.	V_2O_5	a.	The oxidation of ethyne to ethanal
ii.	$\text{TiCl}_4 + \text{Al}(\text{CH}_3)_3$	b.	Polymerization of alkynes
iii.	PdCl_2	c.	Oxidation of SO_2 in the manufacture of H_2SO_4
iv.	Nickel complexes	d.	Polymerization of ethylene

Which of the following is the CORRECT option?

- (A) i – c, ii – a, iii – d, iv – b
 (B) i – c, ii – d, iii – a, iv – b
 (C) i – a, ii – b, iii – c, iv – d
 (D) i – a, ii – c, iii – b, iv – d



74. The correct option representing a Freundlich adsorption isotherm is _____.
 (A) $\frac{x}{m} = kp^{-1}$ (B) $\frac{x}{m} = kp^{0.3}$
 (C) $\frac{x}{m} = kp^{2.5}$ (D) $\frac{x}{m} = kp^{-0.5}$
75. Which of the following is paramagnetic?
 (A) O_2 (B) N_2
 (C) H_2 (D) Li_2
76. A compound 'X' upon reaction with H_2O produces a colourless gas 'Y' with rotten fish smell. Gas 'Y' is absorbed in a solution of $CuSO_4$ to give Cu_3P_2 as one of the products. Predict the compound 'X'.
 (A) $Ca_3(PO_4)_2$ (B) Ca_3P_2
 (C) NH_4Cl (D) As_2O_3
77. The alkane that gives only one mono-chloro product on chlorination with Cl_2 in presence of diffused sunlight is _____.
 (A) isopentane
 (B) 2, 2-dimethylbutane
 (C) neopentane
 (D) n-pentane
78. The hydrolysis reaction that takes place at the slowest rate, among the following is _____.
 (A) 
 (B) 
 (C) $H_3C-CH_2-Cl \xrightarrow{aq. NaOH} H_3C-CH_2-OH$
 (D) $H_2C=CH-CH_2-Cl \xrightarrow{aq. NaOH} H_2C=CH-CH_2OH$
79. An ideal gas expands isothermally from $10^{-3} m^3$ to $10^{-2} m^3$ at 300 K against a constant pressure of $10^5 Nm^{-2}$. The work done on the gas is _____.
 (A) -900 kJ (B) +270 kJ
 (C) -900 J (D) +900 kJ
80. Formula of nickel oxide with metal deficiency defect in its crystal is $Ni_{0.98}O$. The crystal contains Ni^{2+} and Ni^{3+} ions. The fraction of nickel existing as Ni^{2+} ions in the crystal is _____.
 (A) 0.31 (B) 0.96
 (C) 0.04 (D) 0.50
81. Which structure(s) of proteins remain(s) intact during denaturation process?
 (A) Tertiary structure only
 (B) Both secondary and tertiary structures
 (C) Primary structure only
 (D) Secondary structure only
82. The molar solubility of CaF_2 ($K_{sp} = 5.3 \times 10^{-11}$) in 0.1 M solution of NaF will be _____.
 (A) $5.3 \times 10^{-10} mol L^{-1}$
 (B) $5.3 \times 10^{-11} mol L^{-1}$
 (C) $5.3 \times 10^{-8} mol L^{-1}$
 (D) $5.3 \times 10^{-9} mol L^{-1}$
83. Identify the CORRECT formula of 'oleum' from the following.
 (A) $H_2S_2O_8$ (B) $H_2S_2O_7$
 (C) H_2SO_3 (D) H_2SO_4
84. The density of 2 M aqueous solution of NaOH is $1.28 g/cm^3$. The molality of the solution is _____. [Given that molecular mass of NaOH = $40 g mol^{-1}$]
 (A) 1.32 m (B) 1.20 m
 (C) 1.56 m (D) 1.67 m
85. When vapours of a secondary alcohol are passed over heated copper at 573 K, the product formed is _____.
 (A) an alkene (B) a carboxylic acid
 (C) an aldehyde (D) a ketone
86. Crude sodium chloride obtained by crystallization of brine solution does not contain _____.
 (A) $CaSO_4$ (B) $MgSO_4$
 (C) Na_2SO_4 (D) $MgCl_2$
87. The Crystal Field Stabilization Energy (CFSE) for $[CoCl_6]^{4-}$ is $18000 cm^{-1}$. The CFSE for $[CoCl_4]^{2-}$ will be _____.
 (A) $8000 cm^{-1}$ (B) $6000 cm^{-1}$
 (C) $16000 cm^{-1}$ (D) $18000 cm^{-1}$
88. For a reaction, activation energy $E_a = 0$ and the rate constant at 200 K is $1.6 \times 10^6 s^{-1}$. The rate constant at 400 K will be _____. [Given that gas constant, $R = 8.314 J K^{-1} mol^{-1}$]
 (A) $3.2 \times 10^6 s^{-1}$ (B) $3.2 \times 10^4 s^{-1}$
 (C) $1.6 \times 10^6 s^{-1}$ (D) $1.6 \times 10^3 s^{-1}$
89. Match the oxide given in column A with its property given in column B.
- | | Column-A | | Column-B |
|------|-----------|----|------------|
| i. | Na_2O | a. | Neutral |
| ii. | Al_2O_3 | b. | Basic |
| iii. | N_2O | c. | Acidic |
| iv. | Cl_2O_7 | d. | Amphoteric |
- Which of the following is the CORRECT match?
 (A) i - b, ii - d, iii - a, iv - c
 (B) i - b, ii - a, iii - d, iv - c
 (C) i - c, ii - b, iii - a, iv - d
 (D) i - a, ii - d, iii - b, iv - c



90. The most stable carbocation, among the following is _____.

- (A) $\text{CH}_3 - \text{CH}_2 - \overset{\oplus}{\text{CH}}_2$
 (B) $(\text{CH}_3)_3 - \text{C} - \overset{\oplus}{\text{CH}} - \text{CH}_3$
 (C) $\text{CH}_3 - \text{CH}_2 - \overset{\oplus}{\text{CH}} - \text{CH}_2 - \text{CH}_3$
 (D) $\text{CH}_3 - \overset{\oplus}{\text{CH}} - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

91. Which of the following is against the rules of ICBN?

- (A) Generic and specific names should be written starting with small letters.
 (B) Hand written scientific names should be underlined.
 (C) Every species should have a generic name and a specific epithet.
 (D) Scientific names are in Latin and should be italicized.

92. Which of the following conditions will stimulate parathyroid gland to release parathyroid hormone?

- (A) Rise in blood Ca^{+2} levels
 (B) Fall in active Vitamin D levels
 (C) Fall in blood Ca^{+2} levels
 (D) Fall in bone Ca^{+2} levels

93. Select the CORRECT sequence of events.

- (A) Gametogenesis → Gamete transfer → Syngamy → Zygote → Cell differentiation → Cell division (Cleavage) → Organogenesis
 (B) Gametogenesis → Gamete transfer → Syngamy → Zygote → Cell division (Cleavage) → Cell differentiation → Organogenesis
 (C) Gametogenesis → Gamete transfer → Syngamy → Zygote → Cell division → (Cleavage) → Organogenesis → Cell differentiation
 (D) Gametogenesis → Syngamy → Gamete transfer → Zygote → Cell division (Cleavage) → Cell differentiation → Organogenesis

94. The contrasting characteristics generally in a pair used for identification of animals in Taxonomic Key are referred to as:

- (A) Alternate (B) Lead
 (C) Couplet (D) Doublet

95. Which is the most common type of embryo sac in angiosperms?

- (A) Bisporic with two sequential mitotic divisions
 (B) Tetrasporic with one mitotic stage of divisions

- (C) Monosporic with three sequential mitotic divisions
 (D) Monosporic with two sequential mitotic divisions

96. Match the organisms in Column I with habitats in Column II.

	Column-I		Column-II
i.	Halophiles	a.	Hot springs
ii.	Thermoacidophiles	b.	Aquatic environment
iii.	Methanogens	c.	Guts of ruminants
iv.	Cyanobacteria	d.	Salty areas

Select the correct answer from the options given below:

- (A) i – b, ii – d, iii – c, iv – a
 (B) i – d, ii – a, iii – c, iv – b
 (C) i – a, ii – b, iii – c, iv – d
 (D) i – c, ii – d, iii – b, iv – a

97. In which one of the following, both autogamy and geitonogamy are prevented?

- (A) Maize (B) Wheat
 (C) Papaya (D) Castor

98. Which of the following statements is CORRECT?

- (A) Lichens are not good pollution indicators.
 (B) Lichens do not grow in polluted areas.
 (C) Algal component of lichens is called mycobiont.
 (D) Fungal component of lichens is called phycobiont.

99. What type of pollination takes place in *Vallisneria*?

- (A) Male flowers are carried by water currents to female flowers at surface of water.
 (B) Pollination occurs in submerged condition by water.
 (C) Flowers emerge above surface of water and pollination occurs by insects.
 (D) Flowers emerge above water surface and pollen is carried by wind.

100. Which of the following animals are TRUE coelomates with bilateral symmetry?

- (A) Annelids
 (B) Adult Echinoderms
 (C) Aschelminthes
 (D) Platyhelminthes

101. No new follicles develop in the luteal phase of the menstrual cycle because:

- (A) Both FSH and LH levels are low in the luteal phase.
 (B) Follicles do not remain in the ovary after ovulation.
 (C) FSH levels are high in the luteal phase.
 (D) LH levels are high in the luteal phase.



102. Which of the following shows whorled phyllotaxy?
 (A) *Calotropis* (B) Mustard
 (C) China rose (D) *Alstonia*

103. Which of the following hormones is responsible for both the milk ejection reflex and the foetal ejection reflex?
 (A) Relaxin (B) Estrogen
 (C) Prolactin (D) Oxytocin

104. Match the following genera with their respective phylum:

i.	<i>Ophiura</i>	a.	Mollusca
ii.	<i>Physalia</i>	b.	Platyhelminthes
iii.	<i>Pinctada</i>	c.	Echinodermata
iv.	<i>Planaria</i>	d.	Coelenterata

Select the correct option:

- (A) i-c, ii-d, iii-b, iv-a
 (B) i-d, ii-a, iii-c, iv-b
 (C) i-c, ii-d, iii-a, iv-b
 (D) i-a, ii-c, iii-d, iv-b
105. Which of the following is a CORRECT statement?
 (A) IUDs suppress gametogenesis
 (B) IUDs once inserted need not be replaced.
 (C) IUDs are generally inserted by the user herself.
 (D) IUDs increase phagocytosis of sperms in the uterus.
106. Regeneration of damaged growing grass following grazing is largely due to:
 (A) secondary meristem
 (B) lateral meristem
 (C) apical meristem
 (D) intercalary meristem
107. The production of gametes by the parents, the formation of zygotes, the F_1 and F_2 plants, can be understood using
 (A) Wenn diagram
 (B) Pie diagram
 (C) A pyramid diagram
 (D) Punnett square
108. Match the placental types (Column-I) with their examples (Column-II).

	Column-I		Column-II
i.	Basal	p.	Mustard
ii.	Axile	q.	China rose
iii.	Parietal	r.	Dianthus
iv.	Free central	s.	Sunflower

Choose the correct answer from the following option:

- (A) i - r, ii - s, iii - p, iv - q
 (B) i - q, ii - r, iii - s, iv - p

- (C) i - p, ii - q, iii - r, iv - s
 (D) i - s, ii - q, iii - p, iv - r

109. In a marriage between male with blood group A and female with blood group B, the progeny had either blood group AB or B. What could be the possible genotype of parents?

- (A) $I^A i$ (Male) ; $I^B i$ (Female)
 (B) $I^A i$ (Male) ; $I^B I^B$ (Female)
 (C) $I^A I^A$ (Male) ; $I^B i$ (Female)
 (D) $I^A I^A$ (Male) ; $I^B i$ (Female)

110. In the dicot root, the vascular cambium originates from:

- (A) intrafascicular and interfascicular tissue in a ring
 (B) tissue located below the phloem bundles and a portion of pericycle tissue above protoxylem
 (C) cortical region
 (D) parenchyma between endodermis and pericycle

111. In which genetic condition, each cell in the affected person, has three sex chromosomes XXY?

- (A) Turner Syndrome
 (B) Thalassemia
 (C) Klinefelter Syndrome
 (D) Phenylketonuria

112. Bicarpellary ovary with obliquely placed septum is seen in :

- (A) *Sesbania* (B) *Brassica*
 (C) *Aloe* (D) *Solanum*

113. Which of the following sexually transmitted diseases do not specifically affect reproductive organs?

- (A) Chlamydia and AIDS
 (B) Genital warts and Hepatitis-B
 (C) Syphilis and Genital herpes
 (D) AIDS and Hepatitis B

114. Which of the following statements is INCORRECT?

- (A) Female cockroach possesses sixteen ovarioles in the ovaries.
 (B) Cockroaches exhibit mosaic vision with less sensitivity and more resolution.
 (C) A mushroom-shaped gland is present in the 6th - 7th abdominal segments of male cockroach.
 (D) A pair of spermatheca is present in the 6th segment of female cockroach.

115. Which scientist experimentally proved that DNA is the sole genetic material in bacteriophage?

- (A) Jacob and Monod
 (B) Beadle and Tatum
 (C) Messelson and Stahl
 (D) Hershey and Chase



116. Which of the following nucleic acids is present in an organism having 70S ribosomes only?
- (A) Double stranded circular DNA with histone proteins.
- (B) Single stranded DNA with protein coat.
- (C) Double stranded circular naked DNA.
- (D) Double stranded DNA enclosed in nuclear membrane.

117. Match the items of Column I with Column II:

	Column I		Column II
i.	XX-XO method of Sex Determination	a.	Turner's Syndrome
ii.	XX-XY method of Sex Determination	b.	Female heterogametic
iii.	Karyotype-45	c.	Grasshopper
iv.	ZW-ZZ method of Sex Determination	d.	Female homogametic

Select the correct option from the following:

- (A) i – d, ii – b, iii – a, iv – c
- (B) i – b, ii – d, iii – a, iv – c
- (C) i – a, ii – d, iii – b, iv – c
- (D) i – c, ii – d, iii – a, iv – b
118. "Ramachandran plot" is used to confirm the structure of
- (A) DNA
- (B) RNA
- (C) Proteins
- (D) Triacylglycerides
119. In Australia, marsupials and placental mammals have evolved to share many similar characteristics. This type of evolution may be referred to as :
- (A) Adaptive Radiation
- (B) Divergent Evolution
- (C) Cyclical Evolution
- (D) Convergent Evolution
120. Prosthetic groups differ from co-enzymes in that-
- (A) They can serve as co-factors in a number of enzyme - catalyzed reactions.
- (B) They require metal ions for their activity.
- (C) They (prosthetic groups) are tightly bound to apoenzymes.
- (D) Their association with apoenzymes is transient.
121. From the following, identify the CORRECT combination of salient features of Genetic Code
- (A) Degenerate, Non-overlapping, Non-ambiguous
- (B) Universal, Non-ambiguous, Overlapping
- (C) Degenerate, Overlapping, Commaless
- (D) Universal, Ambiguous, Degenerate

122. Which of the following cell organelles is present in the highest number in secretory cells?

- (A) Lysosome
- (B) Mitochondria
- (C) Golgi complex
- (D) Endoplasmic reticulum

123. Match the following RNA polymerases with their transcribed products:

	Column I		Column II
i.	RNA polymerase I	a.	tRNA
ii.	RNA polymerase II	b.	rRNA
iii.	RNA polymerase III	c.	hnRNA

Select the correct option from the following:

- (A) i – c, ii – b, iii – a
- (B) i – a, ii – c, iii – b
- (C) i – a, ii – b, iii – c
- (D) i – b, ii – c, iii – a
124. Which of the following organic compounds is the main constituent of Lecithin?
- (A) Phosphoprotein (B) Arachidonic acid
- (C) Phospholipid (D) Cholesterol
125. Humans have acquired immune system that produces antibodies to neutralize pathogens. Still innate immune system is present at the time of birth because it
- (A) provides passive immunity
- (B) is very specific and uses different macrophages.
- (C) produces memory cells for mounting fast secondary response.
- (D) has natural killer cells which can phagocytose and destroy microbes
126. Non-membranous nucleoplasmic structures in nucleus are the site for active synthesis of
- (A) tRNA (B) protein synthesis
- (C) mRNA (D) rRNA
127. In the process of transcription in eukaryotes, the RNA polymerase I transcribes –
- (A) Precursor of mRNA, hnRNA
- (B) mRNA with additional processing, capping and tailing
- (C) tRNA, 5 S rRNA and snRNAs
- (D) rRNAs – 28 S, 18 S and 5.8 S
128. After meiosis I, the resultant daughter cells have
- (A) four times the amount of DNA in comparison to haploid gamete.
- (B) same amount of DNA as in the parent cell in S phase.
- (C) twice the amount of DNA in comparison to haploid gamete.
- (D) same amount of DNA in comparison to haploid gamete.



129. A population of a species invades a new area. Which of the following condition will lead to Adaptive Radiation?
- Area with many habitats occupied by a large number of species.
 - Area with large number of habitats having very low food supply.
 - Area with a single type of vacant habitat
 - Area with many types of vacant habitats.

130. Match the column I with column II.

	Column I		Column II
a.	Golgi apparatus	i.	Synthesis of protein
b.	Lysosomes	ii.	Trap waste and excretory products
c.	Vacuoles	iii.	Formation of glycoproteins and glycolipids
d.	Ribosomes	iv.	Digesting biomolecules

Choose the right match from options given below:

- a-i, b-ii, c-iv, d-iii
 - a-iii, b-iv, c-ii, d-i
 - a-iv, b-iii, c-i, d-ii
 - a-iii, b-ii, c-iv, d-i
131. What will be the sequence of mRNA produced by the following stretch of DNA?
- 3' ATGCATGCATGCATG 5'
- TEMPLATE STRAND
- 5' TACGTACGTACGTAC 3'
- CODING STRAND
- 3' AUGCAUGCAUGCAUG 3'
 - 3' AUGCAUGCAUGCAUG 5'
 - 5' UACGUACGUACGUAC 3'
 - 3' UACGUACGUACGUAC 5'
132. The main difference between active and passive transport across cell membrane is:
- active transport occurs more rapidly than passive transport.
 - passive transport is non-selective whereas active transport is selective.
 - passive transport requires a concentration gradient across a biological membrane whereas active transport requires energy to move solutes.
 - passive transport is confined to anionic carrier proteins whereas active transport is confined to cationic channel proteins.
133. What initiation and termination factors are involved in transcription in eukaryotes?
- α and σ , respectively
 - σ and ρ , respectively
 - α and β , respectively
 - β and γ , respectively

134. Crossing over takes place between which chromatids and in which stage of the cell cycle?
- Non-sister chromatids of non-homologous chromosomes at Pachytene stage of prophase I.
 - Non-sister chromatids of non-homologous chromosomes at Zygotene stage of prophase I.
 - Non-sister chromatids of homologous chromosomes at Pachytene stage of prophase I.
 - Non-sister chromatids of homologous chromosomes at Zygotene stage of prophase I.
135. Coca alkaloid or cocaine is obtained from:
- Datura*
 - Papaver somniferum*
 - Atropa belladonna*
 - Erythroxylum coca*
136. Which of the following is NOT a feature of active transport of solutes in plants?
- Requires ATP
 - Occurs against concentration gradient
 - Non-selective
 - Occurs through membranes
137. Which of the following statements is CORRECT about the origin and evolution of men?
- Neanderthal men lived in Asia between 1,00,000 and 40,000 years back.
 - Agriculture came around 50,000 years back.
 - The *Dryopithecus* and *Ramapithecus* primates existing 15 million years ago, walked like men.
 - Homo habilis* probably ate meat.
138. What will be the direction of flow of water when a plant cell is placed in a hypotonic solution?
- No flow of water in any direction
 - Water will flow in both directions
 - Water will flow out of the cell
 - Water will flow into the cell
139. Select the INCORRECT statement regarding inbreeding.
- Inbreeding depression cannot be overcome by out-crossing.
 - Inbreeding helps in elimination of deleterious alleles from the population.
 - Inbreeding is necessary to evolve a pure line in any animal.
 - Continued inbreeding reduces fertility and leads to inbreeding depression.



140. In Hatch and Slack pathway, the primary CO_2 acceptor is
 (A) RuBisCO
 (B) Oxaloacetic acid
 (C) Phosphoglyceric acid
 (D) Phosphoenol pyruvate
141. Among the following pairs of microbes, which pair has both the microbes that can be used as biofertilizers?
 (A) *Aspergillus* and *Cyanobacteria*
 (B) *Aspergillus* and *Rhizopus*
 (C) *Rhizobium* and *Rhizopus*
 (D) *Cyanobacteria* and *Rhizobium*
142. Which of the following bacteria reduce nitrate in the soil into nitrogen?
 (A) *Nitrosomonas* (B) *Nitrobacter*
 (C) *Nitrococcus* (D) *Thiobacillus*
143. In mung bean, resistance to yellow mosaic, virus and powdery mildew were brought about by:
 (A) Hybridization and selection
 (B) Mutation breeding
 (C) Biofortification
 (D) Tissue culture
144. One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of:
 (A) Blue and red light
 (B) Violet and green light
 (C) Indigo and green light
 (D) Orange and yellow light
145. Mad cow disease in cattle is caused by an organism which has:
 (A) Free DNA without protein coat
 (B) Inert crystalline structure
 (C) Abnormally folded protein
 (D) Free RNA without protein coat
146. In order to increase the yield of sugarcane crop, which of the following plant growth regulators should be sprayed?
 (A) Cytokinins (B) Ethylene
 (C) Auxins (D) Gibberellins
147. Which of the following statements about methanogens is NOT correct?
 (A) They produce methane gas
 (B) They can be used to produce biogas
 (C) They are found in the rumen of cattle and their excreta
 (D) They grow aerobically and breakdown cellulose-rich food
148. Where is the respiratory electron transport system (ETS) located in plants?
 (A) Intermembrane space
 (B) Mitochondrial matrix
 (C) Outer mitochondrial membrane
 (D) Inner mitochondrial membrane
149. The two antibiotic resistance genes on vector pBR322 are for
 (A) tetracycline and kanamycin
 (B) ampicillin and tetracycline
 (C) ampicillin and chloramphenicol
 (D) chloramphenicol and tetracycline
150. Removal of shoot tips is a very useful technique to boost the production of tea- leaves. This is because:
 (A) Gibberellins delay senescence of leaves.
 (B) Gibberellins prevent bolting and are inactivated.
 (C) Auxins prevent leaf drop at early stages.
 (D) Effect of auxins is removed and growth of lateral buds is enhanced.
151. A biocontrol agent to be a part of an integrated pest management should be
 (A) species-specific and inactive on nontarget organisms
 (B) species-specific and symbiotic
 (C) free living and broad spectrum
 (D) narrow spectrum and symbiotic
152. Match the items given in Column-I with those in Column-II and choose the correct option.
- | | Column – I | | Column – II |
|------|---------------|----|-------------------------|
| i. | Rennin | a. | Vitamin B_{12} |
| ii. | Enterokinase | b. | Facilitated transport |
| iii. | Oxyntic cells | c. | Milk protein |
| iv. | Fructose | d. | Trypsinogen |
- (A) i-c, ii-d, iii-a, iv-b
 (B) i-c, ii-d, iii-b, iv-a
 (C) i-d, ii-c, iii-a, iv-b
 (D) i-d, ii-c, iii-b, iv-a
153. Exploitation of bioresources of a nation by multinational companies without authorization from the concerned country is referred to as
 (A) Biowar (B) Bioweapon
 (C) Biopiracy (D) Bioethics
154. Select the CORRECT statement.
 (A) Expiration is initiated due to contraction of diaphragm.
 (B) Expiration occurs due to external intercostal muscles.
 (C) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration.
 (D) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.



155. Match the following enzymes with their functions:

	Column I		Column II
i.	Restriction endonuclease	a.	Joins the DNA fragments
ii.	Restriction exonuclease	b.	Extends primers on genomic DNA template
iii.	DNA ligase	c.	Cuts DNA at specific position
iv.	<i>Taq</i> polymerase	d.	Removes nucleotides from the ends of DNA

Select the correct option from the following:

- (A) i – b, ii – d, iii – a, iv – c
 (B) i – c, ii – a, iii – d, iv – b
 (C) i – c, ii – d, iii – a, iv – b
 (D) i – d, ii – c, iii – a, iv – b
156. Kwashiorkor disease is due to -
 (A) protein deficiency not accompanied by calorie deficiency
 (B) simultaneous deficiency of proteins and fats
 (C) simultaneous deficiency of proteins and calories
 (D) deficiency of carbohydrates
157. Exploration of molecular, genetic and species level diversity for novel products of economic importance is known as:
 (A) Bioprospecting (B) Biopiracy
 (C) Bioenergetics (D) Bioremediation
158. The maximum volume of air a person can breathe in after a forced expiration is known as:
 (A) Total Lung Capacity
 (B) Expiratory Capacity
 (C) Vital Capacity
 (D) Inspiratory Capacity
159. An enzyme catalysing the removal of nucleotides from ends of DNA is:
 (A) Protease (B) DNA ligase
 (C) Endonuclease (D) Exonuclease
160. A specialised nodal tissue embedded in the lower corner of the right atrium, close to Atrio-ventricular septum, delays the spreading of impulses to heart apex for about 0.1 sec. This delay allows
 (A) the atria to empty completely.
 (B) blood to enter aorta.
 (C) the ventricles to empty completely.
 (D) blood to enter pulmonary arteries.
161. If an agricultural field is liberally irrigated for a prolonged period of time, it is likely to face a problem of:
 (A) Salinity (B) Metal toxicity
 (C) Alkalinity (D) Acidity

162. Match the following cell structure with its characteristic feature:

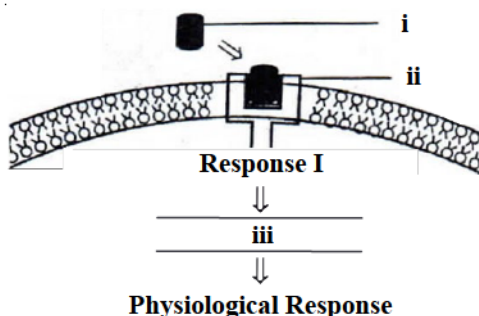
a.	Tight junctions	i.	Cement neighbouring cells together to form sheet
b.	Adhering junctions	ii.	Transmit information through chemical to another cells
c.	Gap junctions	iii.	Establish a barrier to prevent leakage of fluid across epithelial cells
d.	Synaptic junctions	iv.	Cytoplasmic channels to facilitate communication between adjacent cells

Select correct option from the following:

- (A) a-iv, b-iii, c-i, d-ii
 (B) a-ii, b-iv, c-i, d-iii
 (C) a-iv, b-ii, c-i, d-iii
 (D) a-iii, b-i, c-iv, d-ii
163. A selectable marker is used to:
 (A) Mark a gene on a chromosome for isolation using restriction enzyme.
 (B) Help in eliminating the non-transformants, so that the transformants can be regenerated.
 (C) Identify the gene for a desired trait in an alien organism.
 (D) Select a suitable vector for transformation in a specific crop.
164. All the components of the nodal tissue are auto-excitabile. Why does the SA node act as the normal pacemaker?
 (A) SA node has the highest rate of depolarisation.
 (B) SA node has the lowest rate of depolarisation.
 (C) SA node is the only component to generate the threshold potential.
 (D) Only SA node can convey the action potential to the other components.
165. Given below are four statements pertaining to separation of DNA fragments using Gel electrophoresis. Identify the INCORRECT statements.
 i. DNA is negatively charged molecule and so it is loaded on gel towards the anode terminal.
 ii. DNA fragments travel along the surface of the gel whose concentration does not affect movement of DNA.
 iii. Smaller the size of DNA fragment, larger is the distance it travels through it.
 iv. Pure DNA can be visualized directly by exposing to UV radiation.
 Choose correct answer from the options given below:
 (A) i, ii and iv (B) i, iii and iv
 (C) i, ii and iii (D) ii, iii and iv



166. Identify i, ii and iii in the diagrammatic representation of the mechanism of hormone action.



Select the correct option from the following:

- (A) i = Protein Hormone; ii = Cyclic AMP; iii = Hormone-receptor Complex
 (B) i = Steroid Hormone; ii = Hormone-receptor Complex; iii = Protein
 (C) i = Protein Hormone; ii = Receptor; iii = Cyclic AMP
 (D) i = Steroid Hormone; ii = Receptor; iii = Second Messenger
167. In RNAi, the genes are silenced using:
 (A) dsDNA (B) dsRNA
 (C) ss DNA (D) ssRNA
168. Artificial light, extended work-time and reduced sleep-time disrupt the activity of –
 (A) Posterior pituitary gland
 (B) Thymus gland
 (C) Pineal gland
 (D) Adrenal gland
169. Which of the following statements about ozone is CORRECT?
 (A) Stratospheric ozone protects us from UV radiations
 (B) Tropospheric ozone protects us from UV radiations
 (C) Stratospheric ozone is 'bad'
 (D) Tropospheric ozone is 'good'
170. Match the following parts of a nephron with their function:

i.	Descending limb of Henle's loop	a.	Reabsorption of salts only
ii.	Proximal convoluted tubule	b.	Reabsorption of water only
iii.	Ascending limb of Henle's loop	c.	Conditional reabsorption of sodium ions and water
iv.	Distal convoluted tubule	d.	Reabsorption of ions, water and organic nutrients

Select the correct option from the following:

- (A) i-d, ii-a, iii-c, iv-b
 (B) i-a, ii-c, iii-b, iv-d
 (C) i-b, ii-d, iii-a, iv-c
 (D) i-a, ii-d, iii-b, iv-c

171. Carnivorous animals - lions and leopards, occupy the same niche but lions predate mostly larger animals and leopards take smaller ones. This mechanism of competition is referred to as –
 (A) Competitive exclusion
 (B) Character displacement
 (C) Altruism
 (D) Resource partitioning
172. Match the following joints with the bones involved:

i.	Gliding joint	a.	Between carpal and metacarpal of thumb
ii.	Hinge joint	b.	Between atlas and axis
iii.	Pivot joint	c.	Between the Carpals
iv.	Saddle joint	d.	Between Humerus and Ulna

Select the correct option from the following:

- (A) i-a, ii-c, iii-b, iv-d
 (B) i-c, ii-d, iii-b, iv-a
 (C) i-d, ii-a, iii-b, iv-c
 (D) i-d, ii-b, iii-c, iv-a
173. Between which among the following, the relationship is NOT an example of commensalism?
 (A) Female wasp and fig species
 (B) Orchid and the tree on which it grows
 (C) Cattle Egret and grazing cattle
 (D) Sea Anemone and Clown fish
174. Match the items in Column-I with those in Column-II

	Column-I		Column-II
i.	Podocytes	a.	Crystallised oxalates
ii.	Protonephridia	b.	Annelids
iii.	Nephridia	c.	<i>Amphioxus</i>
iv.	Renal calculi	d.	Filtration slits

Select the correct option from the following:

- (A) i-d, ii-b, iii-c, iv-a
 (B) i-c, ii-d, iii-b, iv-a
 (C) i-c, ii-b, iii-d, iv-a
 (D) i-d, ii-c, iii-b, iv-a
175. Western Ghats have a large number of plant and animal species that are not found anywhere else. Which of the following terms will you use to notify such species?
 (A) Keystone (B) Endemic
 (C) Vulnerable (D) Threatened



176. Which of the following diseases is an autoimmune disorder?
- (A) Gout
 - (B) Myasthenia gravis
 - (C) Arthritis
 - (D) Osteoporosis
177. Decline in the population of Indian native fishes due to introduction of *Clarias gariepinus* in river Yamuna can be categorised as
- (A) Alien species invasion
 - (B) Co-extinction
 - (C) Habitat fragmentation
 - (D) Over exploitation
178. Which of the following receptors are specifically responsible for maintenance of balance of body and posture?
- (A) Crista ampullaris and macula
 - (B) Basilar membrane and otoliths
 - (C) Hair cells and organ of Corti
 - (D) Tectorial membrane and macula
179. Which of the following is an innovative remedy for plastic waste?
- (A) Electrostatic precipitator
 - (B) Burning in the absence of oxygen
 - (C) Burying 500 m deep below soil surface
 - (D) Polyblend
180. Which of the following statements is NOT correct?
- (A) In the knee-jerk reflex, stimulus is the stretching of muscle and response is its contraction.
 - (B) An action potential in an axon does not move backward because the segment behind is in a refractory phase.
 - (C) Depolarisation of hair cells of cochlea results in the opening of the mechanically gated potassium-ion channels.
 - (D) Rods are very sensitive and contribute to daylight vision.